Exercise 3
Introduction to SIMULINK

Group 10
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Introduction
Simulink is a graphical simulation environment developed by MathWorks and tightly integrated with MATLAB. It is used for modeling, simulating, and analyzing dynamic systems using block diagrams instead of writing equations line by line.

In Simulink, systems are built by connecting functional blocks (such as integrators, gains, transfer functions, and summation blocks) that represent mathematical operations or physical components. This makes it especially well suited for control systems, signal processing, communication systems, and physical modeling.

Key features of Simulink include:
Visual modeling of continuous, discrete, and hybrid systems
Time-domain simulation with adjustable solvers
Automatic code generation for real-time and embedded applications
Seamless integration with MATLAB for data analysis and visualization
Support for multi-domain modeling (electrical, mechanical, thermal, etc.)

Because of its intuitive block-diagram approach, Simulink is widely used in education, research, and industry to design, test, and validate system behavior before implementation.

3a : Simulate the step response of the transfer-function PT1 G(s) = K$\frac{}{1+Ts}$ with K = 5 and T = 4 sec as differential equation and transfer-function: Display the results in a plot window and export the results to the Matlab workspace. Remember the differential equation of a PT1 is: X = 1/ T (K * u x)

```
% Clear workspace variables, command window, and current figure
clear; clc; clf;

% Gain parameter used in the Simulink model
K = 5;

% Time constant (first case)
T = 4;

% Initial condition of the system state
x0 = -3;

% Run Simulink model with T = 4 and store output in out1
out1 = sim('SimDelay1');

% Change time constant to a new value (second case)
T = 5;

% Run Simulink model again with T = 5 and store output in out2
out2 = sim('SimDelay1');

% Plot the second signal from both simulations versus time
plot(out1.xx.time, out1.xx.signals(2).values, ...  % Output for T
    = 4
```

```
24      out2.xx.time, out2.xx.signals(2).values);      % Output for T
            = 5
25
26  % Label x-axis as time in seconds
27  xlabel('time [sec]');
28
29  % Enable grid for better visualization
30  grid;
31
32  % Add legend to distinguish both responses
33  legend('T=4','T=5');
```

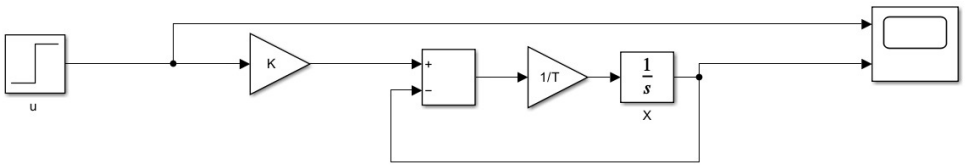


Figure 1: Simulink Model

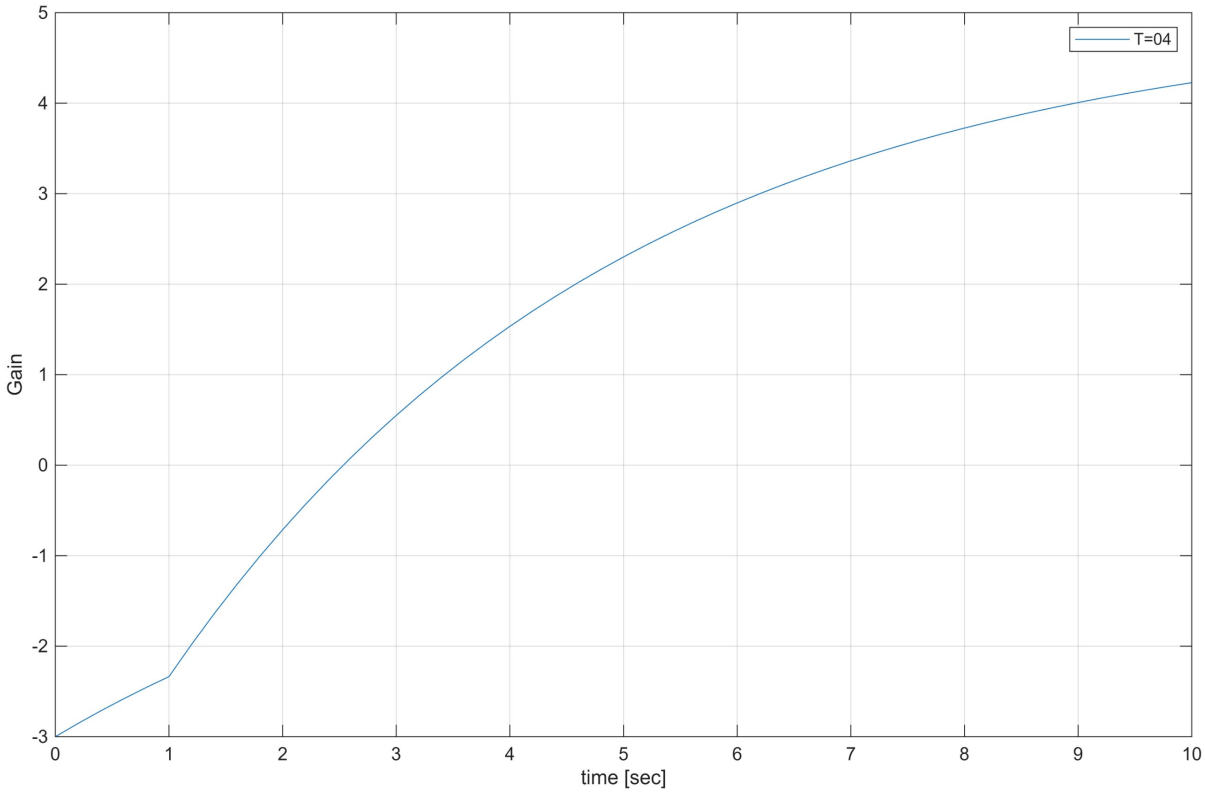


Figure 2: This Simulink model represents a first-order closed-loop dynamic system with negative feedback, driven by a step input. The step block generates the input signal u(t). The input is scaled by a gain K.The summation block subtracts the system output from the scaled input, forming an error signal.The error is multiplied by 1/T, where T is the time constant that controls how fast the system responds.
The integrator 1/s produces the system state/output x(t). The output is fed back negatively, creating stable closed-loop behavior. A scope displays both the input u(t) and the output x(t). In short: the model shows how a first-order system responds to a step input, and how changing the time constant T affects the speed of the response

3b : Build up the model of a one-mass spring system in Simulink.

m x = C x D x + F (t) with

m = 10 kg; C=100 N/m; D=3 N*sec/m;

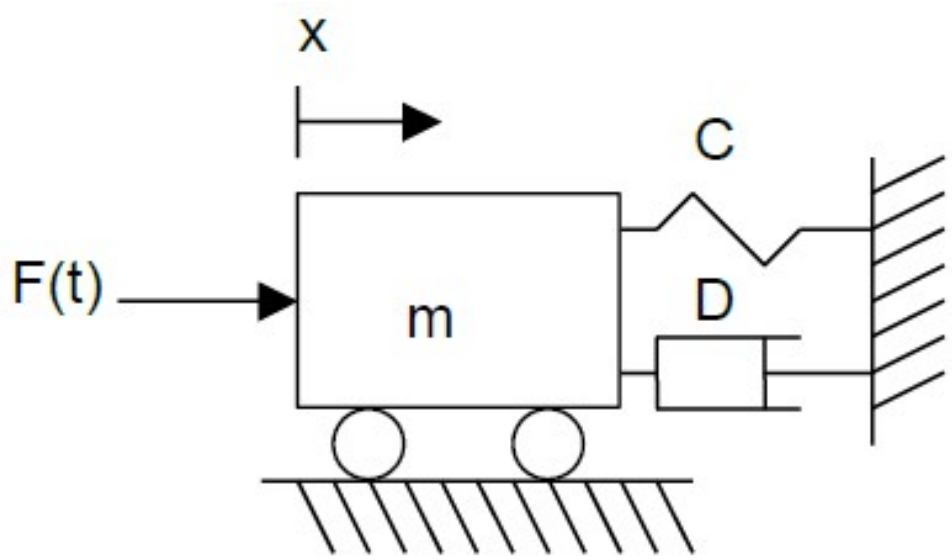


Figure 1:

```
% Mass 1 (kg)
m1 = 100;

% Mass 2 (kg)
m2 = 10;

% Spring constant 1 (N/m)
C1 = 15;

```

```
% Spring constant 2 (N/m)
C2 = 1;

% Damping coefficient 2 (Ns/m)
D2 = 1;

% Run Simulink model and store output
out1 = sim('ex3c');

% Time constant
T = 5;

% Plot the first signal from simulation versus time
plot(out1.xx.time, out1.xx.signals(1).values);

% Label x-axis as time in seconds
xlabel('Time (s)');

% Label y-axis as displacement in meters
ylabel('Displacement (m)');

% Add title to the plot
title('System Response');

% Enable grid for better visualization
grid on;
```

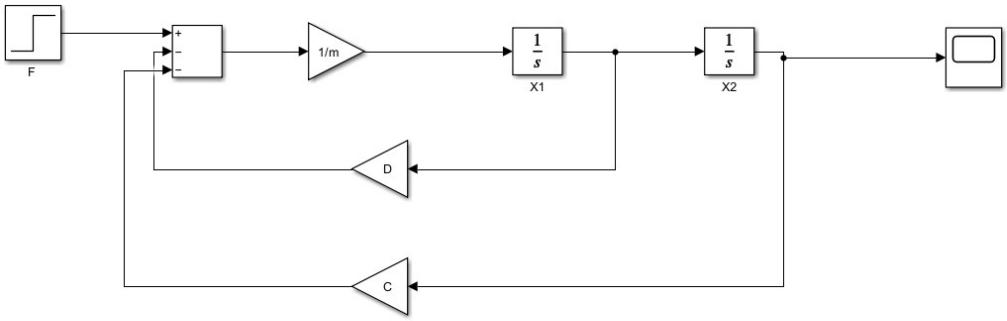


Figure 2: Simulink Model

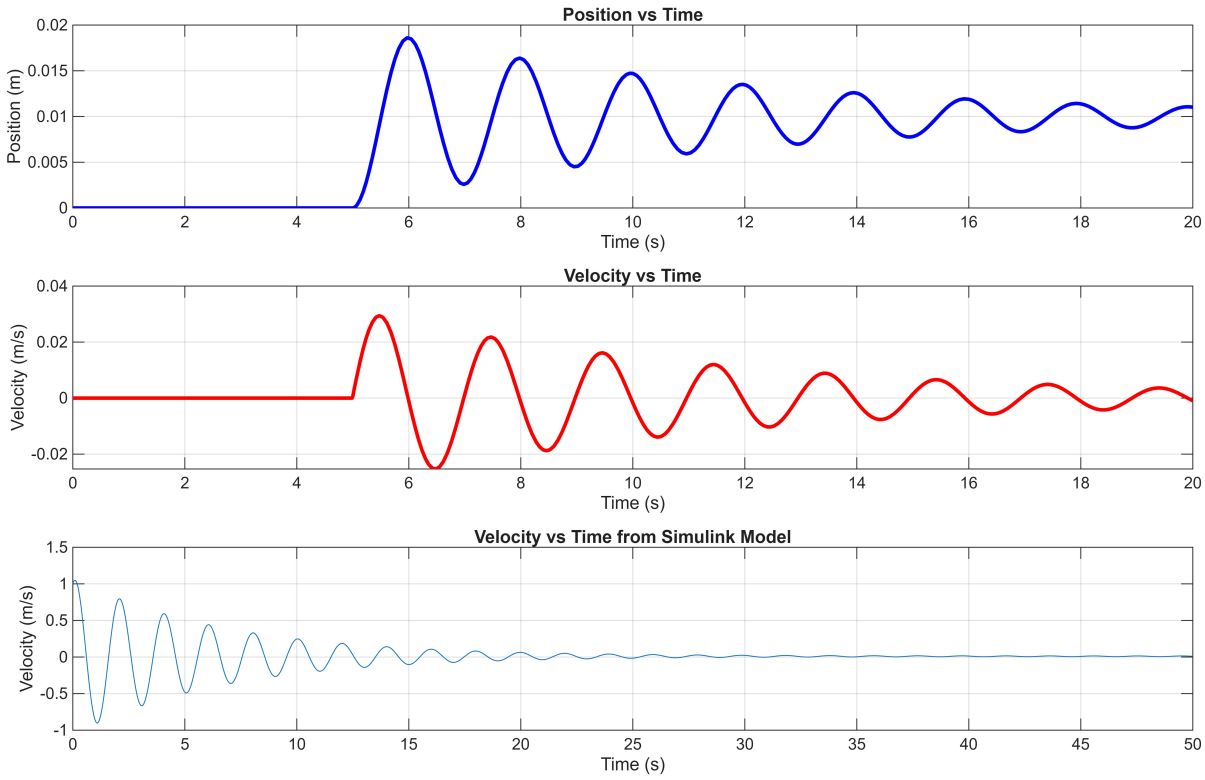


Figure 3: All three plots describe the behavior of a stable, underdamped mechanical system (e.g., mass–spring–damper).
Energy is gradually dissipated due to damping, causing oscillations to decay.
The Simulink result confirms the theoretical/analytical behavior.

3c : Derive the equations for the two mass systems. Optimize the Spring C2 and D2 in a way, that the mass m2=10 kg damps the oscillation of mass m1=100 kg with spring C1=15 N/m using a Simulink simulation model.

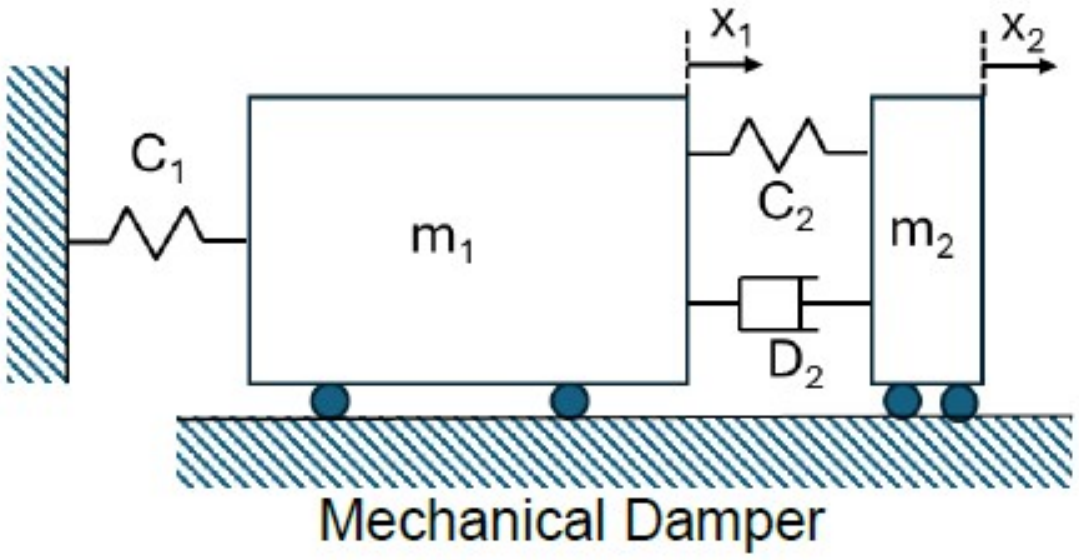


Figure 1:

```
%% Optimization of C2 and D2 for vibration damping

% Fixed system parameters
m1 = 100;    % kg
m2 = 10;     % kg
C1 = 15;     % N/m

% Search ranges for damper parameters
```

```
C2_range = 0.5:0.5:20;      % N/m
D2_range = 0.1:0.2:10;      % Ns/m

bestMetric = Inf;
bestC2 = 0;
bestD2 = 0;

for C2 = C2_range
    for D2 = D2_range

        % Run Simulink model
        out = sim('ex3c');

        % Extract displacement of mass m1
        x1 = out.Group10.signals(1).values;

        % Performance metric: maximum absolute displacement
        metric = max(abs(x1));

        % Store best result
        if metric < bestMetric
            bestMetric = metric;
            bestC2 = C2;
            bestD2 = D2;
        end
    end
end

fprintf('Optimal parameters found:\n');
fprintf('C2 = %.2f N/m\n', bestC2);
fprintf('D2 = %.2f Ns/m\n', bestD2);
fprintf('Minimum peak displacement = %.4f m\n', bestMetric);

%% input Variables
m1 = 100; %kg
m2 = 10;  %kg
C1 = 15;  %N/m
C2 = 1;
%N/m
D2 = 1;
%Ns/m
%% simulation output plot
out1 = sim('ex3c');
T = 5;
plot(out1.Group10.time,out1.Group10.signals(1).values);
```

```
xlabel('Time (s)');
ylabel('Displacement (m)');
title('System Response');
grid on;
```

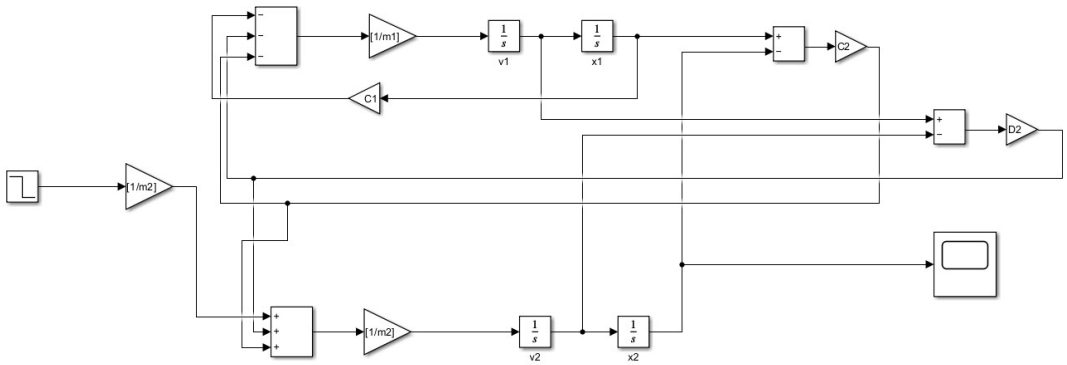


Figure 2: Simulink Model

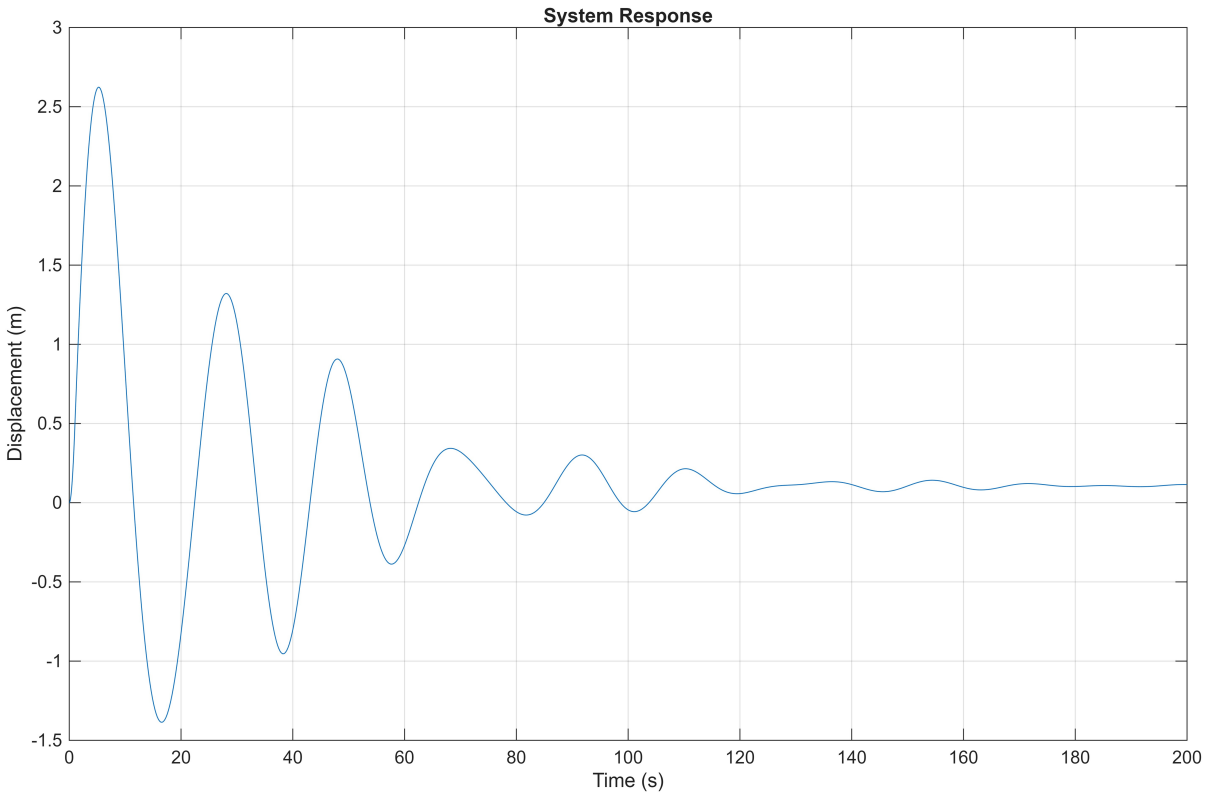


Figure 3: This graph shows the time response of an underdamped system. The displacement initially oscillates with large amplitude. Due to damping, the oscillations gradually decrease over time. The system is stable and eventually settles to a small steady value. No sustained oscillations occur, indicating energy loss in the system.
In summary: the plot represents a damped oscillatory response where the system smoothly settles to equilibrium after initial vibrations.